

Ömer Sayilgan

Curriculum Vitae

Eschenbachstraße 9
Munich
Germany

+49 155 10742256
✉ omersayilgan56@gmail.com



Professional Experience

- 08/2025–
06/2026 **Working Student – Mechatronics & Automation Engineering, Phlair, Munich**
- Developed **TwinCAT 3.1** (Structured Text) logic for an absorber testbench.
 - Calibrated **RS-485** industrial sensors using serial communication tools.
 - Integrated and wired sensors with **Beckhoff I/O terminals** via circuit diagrams.
 - Built data pipelines from PLC to **InfluxDB** using the **PyADS** Python library.
 - Designed real-time monitoring dashboards using **Grafana** and InfluxQL.
 - Deployed local database environments using **Docker, Proxmox**, and Linux networking.
- 08/2023–
05/2024 **Control Systems Engineer, Roketsan Inc., Ankara**
- Designed **PID and Robust Controllers** for a high-precision **missile seeker gimbal** system using classical and modern control techniques.
 - Developed and simulated **dynamic models** in **MATLAB/Simulink** for performance analysis and controller validation.
 - Generated **Embedded C code** from Simulink models and deployed onto **Xilinx** hardware platforms.
 - Conducted **Real-Time (RT) testing** and PID tuning via **xPC Target** and **Hardware-in-the-Loop (HIL)** setups.
 - Developed a **MATLAB GUI** to automate calibration for resolver bias and electrical angle alignment.

Education

- 10/2023–
Present **M.Sc. in Aerospace, Technical University of Munich (TUM), Munich, Germany**
- **Master's Thesis (ongoing):** Fault-Tolerant Control of Propulsive Spacecraft Systems – categorizing spacecraft dynamics and control redundancies, simulating fault scenarios, and evaluating their effect on path planning.
 - **Semester Thesis:** Extended an automated signal integrity check toolbox at the Institute of Flight System Dynamics – developed MATLAB OOP classes, automated Simulink model generation via scripts, and implemented V-Model-based testing with code compliance (Mrails) and test harness structures.
 - **Provisional Grade:** 2.3 — **Credits:** 89 ECTS
 - **Control & GNC:** Advanced Control (2.0), Nonlinear Control (2.0), Optimal Control (2.3), Model Reference Adaptive Control (3.0).
 - **Flight & Spacecraft:** Flight Dynamics, Stability and Control (3.0), System Identification (2.3).
 - **Software & Systems:** Advanced Programming / C++ (1.7), Real-Time Systems (2.0), Introduction to ROS (1.0), MATLAB/Simulink for CAE (2.0).
 - **Labs:** Cyber-Physical Systems Lab – Autonomous Applications (1.9), Flight Control System Lab (1.7).

2017–2023 **B.Sc. Mechanical Engineering**, *Middle East Technical University (METU)*, Ankara

- Final Grade: 3.33 / 4.0 (German Equivalent: **1.7** / “Gut”)
- **Minor Program:** Mechatronics (Robotics, Computer Science, Logic Design)
- **Academic Standing:** Top 10–15% of the department (Rank 25/221 in 2020)
- **Relevant Coursework:** Control Systems, Mechatronic Design, Feedback Systems, Embedded Systems Development
- **Awards:** Multiple “High Honor” and “Honor” roll placements

Skills

Control & GNC	PID, Robust, Nonlinear, Adaptive (MRAC), and Optimal Control (LQR, MPC), Lyapunov Methods, Fault-Tolerant Control, System Identification, 6-DOF Flight Dynamics
Simulation	MATLAB, Simulink, Simulink Code Generation (Embedded Coder), Model-Based Design, SIL/HIL Testing, xPC Target
Programming	C, C++, Embedded C, Python (NumPy, SciPy, PyADS), MATLAB OOP, Git
Automation	TwinCAT 3.1, Beckhoff PLC, Industrial Sensors, RS-485, Serial Protocols
Linux/IT	Ubuntu, Docker, Proxmox, Networking, InfluxDB, Grafana
Robotics	ROS (Robot Operating System), Autonomous Systems

Languages

Turkish	Native
English	Full Professional Proficiency (University Instruction in English)
German	B1

Interests

Hobbies Playing Guitar, Singing, Martial Arts, Travelling

Munich, Germany, August 13, 2026



Ömer Sayilgan